

Body Fluid & Circulation

Essential Substances and Transport Mechanisms:-

All living cells must be continuously supplied with nutrients, oxygen (O_2), and other essential substances. Simultaneously, waste or harmful substances produced must be removed for the healthy functioning of tissues. Efficient mechanisms are therefore necessary for the movement of these substances to and from the cells.

- Simple organisms, such as sponges and coelenterates, circulate water from their surroundings through their body cavities to facilitate the exchange of substances by their cells.
- More complex organisms utilize special fluids within their bodies for transport.
- Blood is the fluid most commonly used for transport by most higher organisms, including humans.
- Lymph (or tissue fluid) is another body fluid that assists in the transport of certain substances.

II. Blood Composition (15.1):-

Blood is classified as a special connective tissue. It consists of a fluid matrix known as plasma and the formed elements.

A. Plasma (15.1.1):-

- Plasma is a straw coloured, viscous fluid.
- It constitutes nearly 55 per cent of the blood.

- Water makes up 90–92 per cent of plasma.
- Proteins contribute 6–8 per cent of plasma content.
- The major proteins are Fibrinogen, Globulins, and Albumins.
 - Fibrinogens are required for the coagulation or clotting of blood.
 - Globulins are primarily involved in the body's defense mechanisms.
 - Albumins help maintain osmotic balance.
- Plasma also contains small amounts of minerals (e.g., Na^+ , Ca^{++} , Mg^{++} , HCO_3^- , Cl^-).
- Substances like glucose, amino acids, and lipids are present in plasma as they are constantly in transit.
- Factors necessary for blood coagulation or clotting are present in plasma in an inactive form.
- Serum is defined as plasma without the clotting factors.

B. Formed Elements (15.1.2):-

Formed elements—erythrocytes, leucocytes, and platelets—collectively make up nearly 45 per cent of the blood.

1. Erythrocytes (Red Blood Cells or RBCs):

- They are the most abundant cells in the blood.
- A healthy adult man typically has 5 million to 5.5 million RBCs per mm^{-3} of blood.
- RBCs are formed in the red bone marrow in adults.

- In most mammals, RBCs are devoid of a nucleus and are biconcave in shape.
- They contain haemoglobin, a red coloured, iron containing complex protein, which gives them their colour and name.
- A healthy individual carries 12–16 gms of haemoglobin in every 100 ml of blood.
- These molecules play a significant role in the transport of respiratory gases.
- RBCs have an average life span of 120 days.
- After 120 days, they are destroyed in the spleen, which is known as the graveyard of RBCs.

2. Leucocytes (White Blood Cells or WBCs):-

- They are colourless because they lack haemoglobin.
- They are nucleated.
- They are relatively few in number, averaging 6000–8000 per mm^{-3} of blood.
- Leucocytes are generally short lived.
- WBCs are categorized into two main groups:
 - Granulocytes: Neutrophils, eosinophils, and basophils.
 - Agranulocytes: Lymphocytes and monocytes.

- Neutrophils are the most abundant WBCs (60–65 per cent).
- Basophils are the least abundant (0.5–1 per cent).
- Neutrophils and monocytes (6–8 per cent) are phagocytic cells that destroy foreign organisms entering the body.
- Basophils secrete substances like histamine, serotonin, and heparin, and are involved in inflammatory reactions.
- Eosinophils (2–3 per cent) resist infections and are also associated with allergic reactions.
- Lymphocytes (20–25 per cent) include 'B' and 'T' forms, both responsible for the body's immune responses.

3. Platelets (Thrombocytes):-

- These are cell fragments produced from specialized cells in the bone marrow called megakaryocytes.
- Normal blood contains 1,50,00 to 3,50,00 platelets per mm^{-3} .
- Platelets release a variety of substances, most of which are involved in the coagulation or clotting of blood.
- A reduction in platelet number can result in clotting disorders, leading to excessive blood loss.

III. Blood Groups (15.1.3):-

Human blood is grouped based on the presence or absence of specific surface antigens. The two most widely used systems

ABO and Rh grouping.

A. ABO Grouping:-

- This grouping is based on the presence or absence of two surface antigens (A and B) on the RBCs. Antigens are chemicals that can induce an immune response.
- The plasma of individuals contains two natural antibodies (proteins produced in response to antigens).
- Matching donor blood with recipient blood is essential before transfusion to prevent severe problems of clumping (destruction of RBC).
- Group A: Has A antigens on RBCs and anti-B antibodies in plasma.
- Group B: Has B antigens on RBCs and anti-A antibodies in plasma.
- Group AB: Has A and B antigens on RBCs and nil antibodies in plasma.
- Group O: Has nil antigens on RBCs and anti-A and anti-B antibodies in plasma.
- Universal Donors: Group 'O' individuals can donate blood to persons with any other blood group.
- Universal Recipients: Group 'AB' individuals can accept blood from persons with AB and all other groups.

B. Rh Grouping:-

- The Rh antigen, similar to one found in Rhesus monkeys, is

is present on the surface of RBCs in nearly 80 per cent of humans.

- Individuals with the antigen are called Rh positive (Rh+ve); those without it are Rh negative (Rh-ve).

- An Rh-ve person, if exposed to Rh+ve blood, will form specific antibodies against the Rh antigens. Therefore, Rh groups must be matched during transfusions.

- Erythroblastosis Foetalis: This condition results from a special case of Rh incompatibility.

- It occurs when an Rh-ve mother is pregnant with an Rh+ve foetus.

- During the first pregnancy, the foetal and maternal bloods are typically separated by the placenta.

- However, during the delivery of the first child, the mother may be exposed to small amounts of foetal Rh+ve blood, prompting her to prepare anti-Rh antibodies.

- In subsequent pregnancies, these maternal Rh antibodies can leak into the blood of an Rh+ve foetus and destroy its RBCs.

- This condition can be fatal or cause severe anaemia and jaundice in the baby.

- Erythroblastosis foetalis can be avoided by administering anti-Rh antibodies to the mother immediately following the delivery of the first child.

IV. Coagulation of Blood (15.1.4):-

Clotting is a mechanism exhibited by blood in response to an injury or trauma to prevent excessive blood loss.

- A dark reddish brown scum, called a clot or coagulum, is formed at the site of injury.
- The clot is mainly composed of a network of threads called fibrins that trap dead and damaged formed elements.
- Fibrins are formed by the conversion of inactive fibrinogens (present in plasma) by the enzyme thrombin.
- Thrombins are formed from another inactive plasma substance called prothrombin.
- An enzyme complex known as thrombokinase is required for the conversion of prothrombin to thrombin.
- The formation of thrombokinase involves a series of linked enzymic reactions, known as a cascade process, involving numerous factors present in the plasma in an inactive state.
- An injury or trauma stimulates platelets in the blood to release factors that activate the coagulation mechanism. Factors released by tissues at the injury site can also initiate coagulation.
- Calcium ions (Ca^{++}) play a crucial role in clotting.

V. Lymph (Tissue Fluid) (15.2):-

- As blood flows through the capillaries in tissues, water and many small water soluble substances move out into the spaces between the tissue cells.
- Larger proteins and most formed elements remain in the

in the blood vessels.

- This fluid released out is called the interstitial fluid or tissue fluid.
- Tissue fluid has the same mineral distribution as plasma.
- Exchange of nutrients, gases, etc., between the blood and the cells always happens through this interstitial fluid.
- An elaborate network of vessels known as the lymphatic system collects this fluid and drains it back to the major veins.
- The fluid within the lymphatic system is called lymph.
- Lymph is a colourless fluid.
- It contains specialized lymphocytes, which are responsible for the immune responses of the body.
- Lymph is an important carrier for hormones and nutrients.
- Fats are absorbed through lymph in the lacteals present in the intestinal villi.
- Lymph is similar to blood, with the exception of its formed element and protein content.

VI. Circulatory Pathways (15.3):-

Circulatory patterns are classified as either open or closed.

- **Open Circulatory System:** Found in arthropods and molluscs. Blood is pumped by the heart through large vessels into open spaces or body cavities called sinuses.

- **Closed Circulatory System:** Found in annelids and chordates. Blood pumped by the heart is always circulated through a closed network of blood vessels. This pattern is considered more advantageous as the flow of fluid can be precisely regulated.

Vertebrate Circulatory Hearts:-

All vertebrates possess a muscular chambered heart.

- **Fishes:** Have a 2-chambered heart (one atrium and one ventricle). They exhibit single circulation where the heart pumps deoxygenated blood, which is then oxygenated by the gills before being supplied to the body.
- **Amphibians and most Reptiles (except crocodiles):** Have a 3-chambered heart (two atria and a single ventricle). They undergo incomplete double circulation because the oxygenated blood (from the left atrium) and deoxygenated blood (from the right atrium) mix in the single ventricle before being pumped out.
- **Crocodiles, Birds, and Mammals:** Possess a 4-chambered heart (two atria and two ventricles). They have double circulation with two separate circulatory pathways, ensuring oxygenated and deoxygenated blood pass through the ventricles without mixing.

VII. Human Circulatory System (15.3.1)

The human circulatory system, or blood vascular system, consists of blood, a network of closed branching blood vessels, and a muscular chambered heart.

Heart Structure

- The heart is a mesodermally derived organ situated in the thoracic cavity, located between the two lungs and slightly tilted to the left.
- It is the size of a clenched fist.
- It is protected by the pericardium, a double-walled membranous bag enclosing the pericardial fluid.
- The heart has four chambers: two relatively small upper chambers called atria and two larger lower chambers called ventricles.
- The right and left atria are separated by the thin, muscular inter-atrial septum.
- The left and right ventricles are separated by the thick-walled inter-ventricular septum.
- The atrium and ventricle on the same side are separated by a thick fibrous tissue, the atrio-ventricular septum.
- Valves: These structures guard the openings between chambers and vessels, allowing blood flow in only one direction.
 - The opening between the right atrium and the right ventricle is guarded by the tricuspid valve (formed of three muscular flaps/cusps).
 - The opening between the left atrium and the left ventricle is guarded by the bicuspid or mitral valve.
 - Semilunar valves guard the openings where the right

- The entire heart is constructed of cardiac muscles. The walls of the ventricles are considerably thicker than those of the atria.

Nodal Tissue and Pacemaker

- A specialized cardiac musculature called nodal tissue is distributed throughout the heart.
- The sino-atrial node (SAN) is a patch of this tissue found in the right upper corner of the right atrium.
- The atrio-ventricular node (AVN) is a mass of tissue located in the lower left corner of the right atrium, close to the atrio-ventricular septum.
- The atrio-ventricular bundle (AV bundle) continues from the AVN, passes through the atrio-ventricular septa, and divides immediately into right and left bundles on the inter-ventricular septum.
- These bundles give rise to minute fibres called Purkinje fibres throughout the ventricular musculature.
- The nodal musculature is autoexcitable, meaning it can generate action potentials without external stimuli.
- The SAN generates the maximum number of action potentials (70–75 per minute) and initiates and maintains the heart's rhythmic contractile activity.
- The SAN is therefore called the pacemaker.
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VIII. Cardiac Cycle (15.3.2)

The sequential events in the heart that are cyclically repeated constitute the cardiac cycle, consisting of the systole (contraction) and diastole (relaxation) of both the atria and ventricles. A healthy person performs approximately 72 cardiac cycles per minute, meaning the duration of one cycle is 0.8 seconds.

- **Joint Diastole:** The cycle begins with all four chambers in a relaxed state. The tricuspid and bicuspid valves are open, while the semilunar valves are closed. Blood flows into the ventricles from the pulmonary veins (to the left atrium/ventricle) and vena cava (to the right atrium/ventricle).
- **Atrial Systole:** The SAN generates an action potential, causing simultaneous contraction of both atria. This increases the flow of blood into the ventricles by about 30 per cent.
- **Ventricular Systole:** The action potential reaches the ventricles via the AVN, AV bundle, and bundle of His, initiating ventricular contraction. Atrial relaxation (diastole) coincides with this event.
 - Ventricular systole increases ventricular pressure,

leading to the closure of the tricuspid and bicuspid valves, preventing backflow into the atria.

- As pressure rises further, the semilunar valves are forced open, ejecting blood into the pulmonary artery and the aorta.
- Ventricular Diastole: The ventricles relax, and pressure falls, causing the closure of the semilunar valves (preventing backflow into the ventricles).
- As ventricular pressure drops further, the pressure exerted by the blood in the atria pushes the tricuspid and bicuspid valves open, and blood flows freely into the ventricles, returning the heart to joint diastole.

Cardiac Output and Heart Sounds

- Stroke Volume: During each cardiac cycle, each ventricle pumps out approximately 70 mL of blood.
- Cardiac Output: This is the volume of blood pumped out by each ventricle per minute. It is calculated as Stroke Volume multiplied by the Heart Rate.
- The average cardiac output in a healthy individual is 5000 mL or 5 litres. The body can alter the stroke volume and heart rate to adjust cardiac output (e.g., higher in an athlete).
- Heart Sounds: Two main sounds are produced per cycle, used for clinical diagnostic significance.
 - The first heart sound (lub) is associated with the closure of the tricuspid and bicuspid valves.

- The second heart sound (dub) is associated with the closure of the semilunar valves.

IX. Electrocardiogram (ECG) (15.3.3)

The electrocardiogram (ECG) is a graphical representation of the electrical activity of the heart during a cardiac cycle, recorded using a machine called an electro-cardiograph.

- To obtain a standard ECG, a patient is connected with three electrical leads (one to each wrist and one to the left ankle). Multiple leads attached to the chest provide a detailed evaluation.
- The ECG is composed of identified peaks (P to T) corresponding to specific electrical activities:
 - P-wave: Represents the electrical excitation (or depolarisation) of the atria, which triggers the contraction of both atria.
 - QRS complex: Represents the depolarisation of the ventricles, initiating ventricular contraction, which starts shortly after Q and marks the beginning of the systole.
 - T-wave: Represents the return of the ventricles from the excited to the normal state (repolarisation); the end of the T-wave marks the end of systole.
- The heart beat rate can be determined by counting the number of QRS complexes occurring in a specific time period.
- Any deviation from the typical ECG shape indicates a possible abnormality or disease, giving it great clinical significance.

X. Double Circulation (15.4)

Humans have a complete double circulation consisting of two separate pathways: pulmonary and systemic. Blood flows through fixed routes in vessels.

- Blood Vessel Structure: Arteries and veins basically consist of three layers:

- Tunica intima: The inner lining of squamous endothelium.

- Tunica media: The middle layer composed of smooth muscle and elastic fibres (this layer is comparatively thin in veins).

- Tunica externa: The external layer made of fibrous connective tissue containing collagen fibres.

1. Pulmonary Circulation

- The pathway begins when the right ventricle pumps deoxygenated blood into the pulmonary artery.

- The blood travels to the lungs where it is oxygenated.

- Oxygenated blood is then carried by the pulmonary veins into the left atrium.

2. Systemic Circulation

- The pathway begins when the left ventricle pumps oxygenated blood into the aorta.

- This blood is distributed by a network of arteries, arterioles, and capillaries to the body tissues.

- Deoxygenated blood is collected from the tissues by a system of venules, veins, and the vena cava, which empty the blood into the right atrium.
- The systemic circulation supplies O_2 and essential substances to the tissues and removes carbon dioxide (CO_2) and other harmful substances for elimination.

Special Circulations

- **Hepatic Portal System:** A unique vascular connection exists between the digestive tract and the liver. The hepatic portal vein carries blood from the intestine to the liver before the blood enters the systemic circulation.
- **Coronary System:** A specific system of blood vessels present in the body exclusively for the circulation of blood to and from the cardiac musculature (heart muscle).

XI. Regulation of Cardiac Activity (15.5)

The heart is called myogenic because its normal activities are regulated intrinsically (auto regulated) by the specialized nodal tissue muscles.

- A special neural center in the medulla oblongata can modulate cardiac function via the Autonomic Nervous System (ANS).
- Sympathetic nerves (part of the ANS) increase the heart rate, increase the strength of ventricular contraction, and consequently increase the cardiac output.
- Parasympathetic neural signals (another component of the ANS) decrease the heart rate, decrease the speed of action

potential conduction, and thereby decrease the cardiac output.

- Adrenal medullary hormones also have the ability to increase the cardiac output.

XII. Disorders of the Circulatory System (15.6)

1. High Blood Pressure (Hypertension):

- The term refers to blood pressure higher than the normal 120/80 mm Hg.
- 120 mm Hg represents the systolic (pumping) pressure, and 80 mm Hg is the diastolic (resting) pressure.
- Hypertension is indicated if repeated checks show blood pressure at 140/90 or higher.
- High blood pressure contributes to heart diseases and affects vital organs such as the brain and kidney.

2. Coronary Artery Disease (CAD) / Atherosclerosis:

- CAD affects the vessels responsible for supplying blood to the heart muscle.
- It is caused by deposits of calcium, fat, cholesterol, and fibrous tissues, which result in the narrowing of the artery lumen.

3. Angina (Angina Pectoris):

- This is a symptom defined as acute chest pain that appears when the heart muscle is not receiving sufficient oxygen.

- Angina is caused by conditions that impede blood flow.
- It can occur in men and women of any age, but is more prevalent among the middle-aged and elderly.

4. Heart Failure:

- This describes the state of the heart when it is unable to pump blood effectively enough to satisfy the needs of the body.
- It is sometimes called congestive heart failure because lung congestion is one of its primary symptoms.
- Crucially, heart failure is not the same as cardiac arrest (when the heart stops beating) or a heart attack (when the heart muscle is suddenly damaged by inadequate blood supply)